

information about the keyframe, then all subsequent frames to build the final image. More keyframes mean larger files, but lesser seek time. Also, frames further away from the keyframe are more susceptible to low quality, since each time only an approximate difference is stored. Thus, you should keep the keyframe interval to as low as possible.

Avoid outdated formats

Microsoft Video 1, Radius Cinepak, and Indeo 3.2 are all outdated formats. Video 1 cannot always render perfect frames, since it can assign only eight colours to every 16 pixels. Radius Cinepak drastically affects the picture quality with its motion prediction function and also creates very large files. Indeo 3.2 can cause colour bleeding and dithering. Though Indeo 5 is much better than Indeo 4, the latter is a good codec for AVI files if you plan to distribute the video to a wide audience; Indeo 5 cannot be decoded on a Macintosh. Also, avoid proprietary formats such as MJPEG, VCR1, VCR2, ASV1, and WNV1—codecs for these are not easy to find and are not free.

Recording sound

If your hardware allows it, plan beforehand whether you want to record sound along with the video itself or from an external source. An external source usually

means much better quality, but can also prove to be sluggish and is a hassle.

Since one audio file is created for every video, you have to keep track of double the number of files in post-production. Also, synchronising audio and video involves more time and has to be done perfectly. However, if you capture video and audio simultaneously and the system is slow, it may be susceptible to dropping frames as it has to process the video and the audio at the same time.

Audio levels

The capture volume should be high enough for clear sound. Softer is preferable over louder, as long as the sound is audible. 16-bit audio is better for capturing, since it has much more precision than 8-bit.

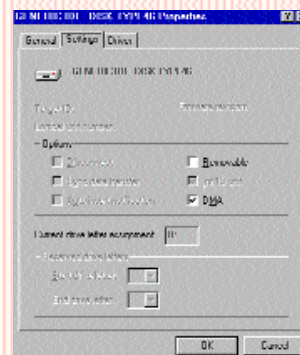
Editing with speed

Capturing and editing video is much better with loads of computing power and disk space, but even on the best machines, processing video is a very slow process. Here are a few things you can do to speed it up

Enable DMA for IDE

If you have Windows 98 or later, enable DMA for the hard disk. Open **Control Panel > System > Device Manager** and double-click each disk under Disk Drives. Check the DMA box under Settings and click OK. Older drives may not

support DMA, so the next time you reboot, some of these boxes may be unchecked.



Enable DMA for your hard disk for faster access

Windows 98 or better

Windows 98 has a much more efficient disk subsystem than Windows 95, which allows better writing to the hard disk. This can significantly improve capture quality and editing speed. Windows 2000 is preferred over any other Windows version. Windows 95, 98 and Me will not allow AVI files over 2 GB. If you use Windows 2000 or XP, there is no such limitation, but do turn off some of the operating system's memory-hogging features if you use XP.

Use faster partitions

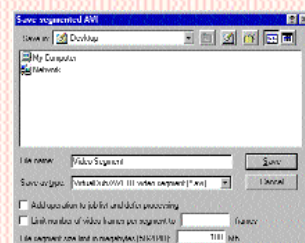
Files stored at the start of a disk are faster to access. Make sure that the video is captured to a partition which is at the start of the disk and which is not already too full.

Note that the position of

the actual data matters, not the size or position of the partition. For a disk split in half, data at the end of the first partition (when the drive is nearly full) will be read at the same speed as data at the start of the second.

Sum of all parts

When editing a large video file, it may be a good idea to break the file into smaller bits. You will probably be applying filters and recompressing the video after editing. Once you finish cleaning up each bit, you can join them together again and recompress and apply the filters at the same time.



VirtualDub can automatically cut video into smaller parts

Mixing filters

If you are applying multiple effects to the video, arrange them in the best possible order if the software allows you to do so. Doing so can cut the conversion process by half or even more. For example, if you apply a filter to smoothen out the video and another one to resize it, place the resize filter first—the smoothening filter will have



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